Innovation & Regulation

Introduction I

Dr. M. K. (Konrad) Borowicz

Plan for today

- Information about the course
- Defining innovation
- Defining regulation
- Why regulate?

What this course is about

- This course explores the interaction between innovation and regulation in the field of data science.
- It draws mainly on economic theories of innovation and regulation, and focuses on data driven innovation.
- It discusses theories of innovation and regulation in general and applies them in specific fields, such as agriculture and finance.

Course objectives

Upon completion of this course, you will have obtained a basic understanding of the fundamental theories and principles underpinning the regulation of innovation, and the interplay between regulation and innovation in general, and innovation driven by (big) data in particular.

You will be able to:

1. identify real-life examples of (data driven) innovation and evaluate the potential social and economic impact of such innovation;
2. identify and understand the role of economic, social and rights-based considerations as grounds for regulation of innovative technologies;
3. identify different types of regulatory strategies and regimes applicable to innovation regulation, including risk-based strategies;
4. rely on your knowledge of regulatory regimes to inform decisions in engineering design;
5. recognize the importance of regulatory regimes for innovation and engineering design more generally;
6. apply principles of responsible innovation in engineering design.

Relationship to other course

- Law and Data Science
  - Introduced you to some basic legal regimes and concepts (e.g. property rights). We will talk about some of the same regimes, which are pf particular importance for data science studnets such as IP and data protection/transfer but we will focus mostly on economic rationales and more detailed rules
- Data Science Ethics
  - We focus on economic theories, but there will be some discussion of other social and ethical considerations

Set up & team



Tjaša Petročnik



Dr. Sunimal Mendis

The course has 6 modules:

- Module 1: Introduction
- Module 2:Helath and data sharing
- Module 3: Data and regulation of competition Module 4: IP and database protection
- Module 5: AI and data regulation
- Module 6: Financial and sustainability data



Dr. Chiara Gallese



Merve Öner Kabadayi

Assignments/exam

- At the end of the course there will be a written exam, which will account for the remaining 100% of the **Final Grade**. This written exam will consist of open-ended questions from all of the modules.

Exam: November 4
Resit: January

What is innovation?

- Many different interpretations
- Butenko and Larouche (2015)
    1. A novel idea or invention;
    2. Its diffusion or adoption by users, customers or citizens-as the case may be;
    3. A positive social impact, in the form of the increase in welfare or a contribution to the achievement of public policy aims, as the case may be and depending on analytical perspective.



Literature guide for this course

- Law and economics literature on innovation
  - deals with the first two elements
  - assumes that innovation is good
- Law and technology literature on innovation
  - Deals with first and third element
  - not particularly concerned with conditions of diffusion
  - interested in 'good' innovation

Some examples – do they meet the criteria of the definition?

- Precision agriculture
- Health wearables
- Social media algorithms
- Smart energy grid
- Facial recognition technology
- High-frequency trading algorithms

What is regulation?

- Broad definition of regulation
  - "any instrument (legal or non-legal, governmental or non-governmental) … that is designed to channel behavior" (Brownsword and Somsen, 2009)
  - "intentional influencing of someone's or somethings behavior" (Black 2005)
- Clearly broader than 'law'
- The term governance is also sometimes used

Forms of regulation

- **European laws** (adopted by European Council and European Parliament, e.g. GDPR);
- **National legislation** (e.g. The California Consumer Privacy Act);
- **Executive orders/rules** (e.g. Executive Order on Improving the Nation's Cybersecurity);
- **Rules of regulatory agencies** (e.g. FTC's Standards for Safeguarding Customer Information);
- **Legal administrative decisions** (e.g. decision of Dutch data protection authority/National competition authority).
- **European and national case law** (e.g. Schrems II case)

Forms of regulation

- **Standards** (adopted by international, European and national standardization bodies, e.g., ISO/IEC 27001);
- **Non binding rules such as guidelines** (e.g.. OECD Guidelines on the Protection of Privacy and Transborder Flows of Personal Data);
- **Self regulation** (private standards, e.g., Visa Core Rules);
- **Technology** (Code is Law, Lessig 1999, e.g., blockchain)

Forms of regulation

- Public and private
- Binding and non-binding
- International, European, national
- Technology

General rules vs specific rules

- General (apply across industries)
  - Data protection laws
  - Competition law
  - IP law
  - Contract law
  - Consumer law

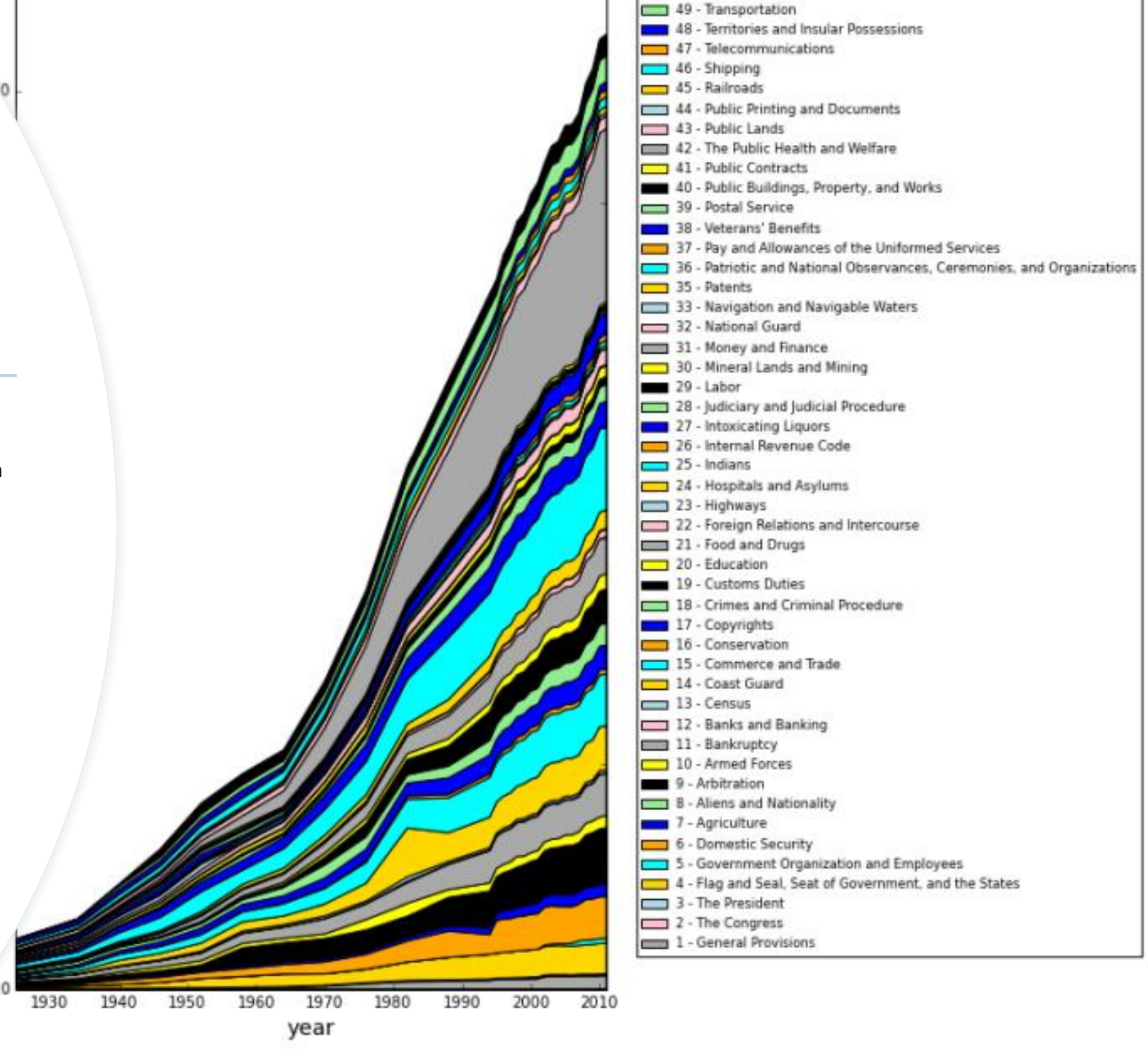
General vs specific rules

- Specific (apply to specific industries)
  - Financial
    - (e.g., Markets in Financial Instruments Directive 2)
  - Healthcare
    - (e.g., Health Insurance Portability and Accountability Act)
  - Education
    - (e.g., The Family Educational Rights and Privacy Act)
  - Energy
    - (e.g., European Network Codes)

Where can you find them?

- https://europa.eu/
- Case law
- Books
- Websites ministries and regulators etc.

- Stacked area graph of the size of the U.S. Code, color-coded by title and measured in the number of words.
- Consistent with the popular conception of federal laws, the size of the U.S. Code has grown continuously since 1926. Moreover, the rate of growth is increasing.
- Source: Li et al. (2014), *Law is Code: A Software Engineering Approach to Analyzing the United States Code*



Why regulate?

Property rights stimulate innovation

- Through knowledge-spillovers
- This is why in this course we start with IP!

Regulation needed:

- In case of market failures;
- Rights based and social rationales for regulating;

Property rights defined

- Cooter and Ulen: property is

  "A bundle of legal rights over resources that the owner is free to exercise and whose exercise is protected from interference by others"

- Possessory rights and rights of transfer
- Property rights are not absolute
  - Governments can interfere when this is in the public interest and subject to the conditions provided for by law

Economic rationale for property rights

1. Avoidance of dispute and of efforts to protect or to take things
2. Incentives to work (create, innovate etc.)
3. Incentives to maintain and improve things (innovate etc.)
4. Incentives to transfer things (ability to trade)

Incentives to work (create, innovate etc.)

- If your output is not protected you have no incentives to work, or work more or less optimal from the point of view of social welfare (too little OR too much)

Hours of work	Output	Utility from output	Disutility from work	Social welfare
0	0	0	0	0
1	1	10	6	4
2	2	18	13	5
3	3	24	22	2

Incentives to work (create, innovate etc.)

- Rights in one's output
  - the socially optimal outcome is achievable under property rights where each individual has rights in his own output;
  - if each individual has property rights in the output he produces, he will work the socially optimal amount and social welfare will be maximized

- Supervisor's rights in one's output
  - the socially optimal outcome is also achievable under property rights where a supervisory entity has rights in an individual's output if the supervisor can monitor individual's work

Incentives to work

- Incentives and salary structure:
  - In real life (other than farm life…) incentives and salary structure established by supervisory entities where agents are employed

- Consequences of breakdown of property rights regime
  - During war there is no productive activity

Incentives to maintain and improve things

- Improvements
  - When things may be taken from individuals, they will not benefit from improving them (if a machine would be taken from an individual, they will not have an incentive to maintain it well)
- Improvements and transfer rights
  - Socially optimal improvement in durable things will be promoted not only by possessory rights, but also by rights to transfer things.
  - Possessors will often transfer things by selling them and sales prices will generally reflect improvements made in things because present owners will anticipate this, they will be the case.

Incentives to transfer things (ability to trade)

- Trade
  - the direct advantage of transfer of goods, that is, of trade, is that it raises the utilities of those who engage in it.
  - this may be because the preferences of individuals differ (hence by transfer we increase their well-being and utility)

- Specialization
  - Efficient methods of production take advantage of specialization and of agglomerations of individuals who devote themselves to making just one or several related goods (this argument goes back to Adam Smith)

Intellectual property rights

- From an innovation policy standpoint, the concern about appropriation is particularly relevant because it could reduce incentives to invest in R&D
- R&D generates knowledge spillovers, which is good!
- But you don’t want overinvestment, which would benefit only firms but not society as a whole
- IP rights are an (imperfect) mechanism to promote R&D investment at adequate levels

How should property rights be allocated to achieve efficiency?

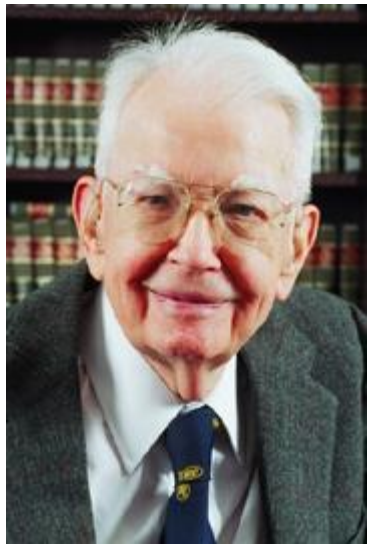
- But first what do we mean by efficiency?
- Efficient allocation
  - Pareto: makes everyone better-off
  - Kaldor-Hicks:increases the total value achieved by everyone in society even if it makes some people worse-off
- So how should property rights be allocated to achieve efficiency?
- Coase's surprising answer: **it doesn't matter**

- (Under certain conditions)

The Coase Theorem

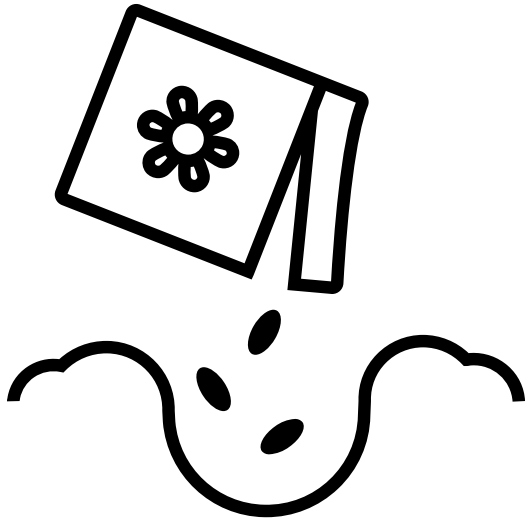
- Ronald Coase (1960), *"The Problem of Social Cost"*

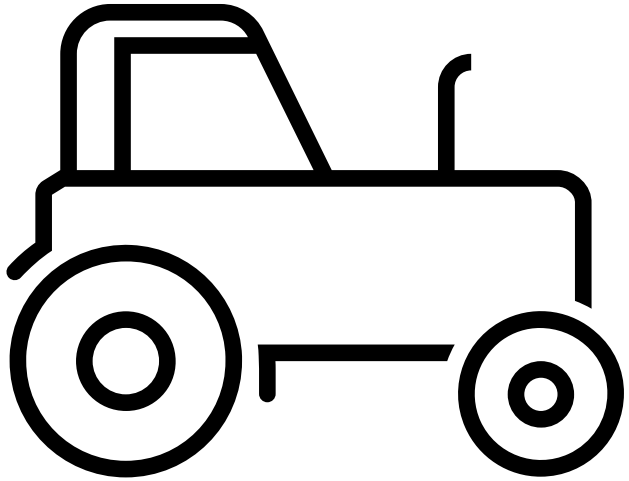
- **In the absence of transaction costs, if property rights are well-defined and tradable, voluntary negotiations will lead to efficiency.**
  - It doesn't matter how rights are allocated initially…
  - …because if they're allocated inefficiently at first, they can always be sold/traded…
  - so the allocation will end up efficient anyway



Ronald Coase
1910-2013

- Initial allocation **does** matter for **distribution**, though
  - And if there are **transaction costs**, may matter for efficiency too







Why regulate?

Baldwin et al.

General market failures

- Monopolies, natural monopolies;
- Externalities;
- Information adequacies;
- Anti-competitive behavior
- Unequal bargaining power;
- Public goods;
- Continuity and availability services;

Data-market specific

Monopolies

A single seller occupies the entire market.

The product sold is unique in the sense that there is no substitute sufficiently close for consumers to turn to.

Substantial barriers restrict entry by other firms into the industry, and exit is difficult.

Monopolies

Effects of monopoly

- reduced output,
- higher prices, and
- transfer of income from consumers to producers.

Policy response: competition law

Natural monopoly

Data markets

- E.g., reduction of search results

Externalities (or spillovers)

- General: Price does not accurately reflect social cost
  - Example: pollution
- Data-specific:
  - Privacy externality

    users suffering privacy loss for which they are not compensated
  - Information externality

    users do not internalize the cost of their information choices to other users
  - Trust externality

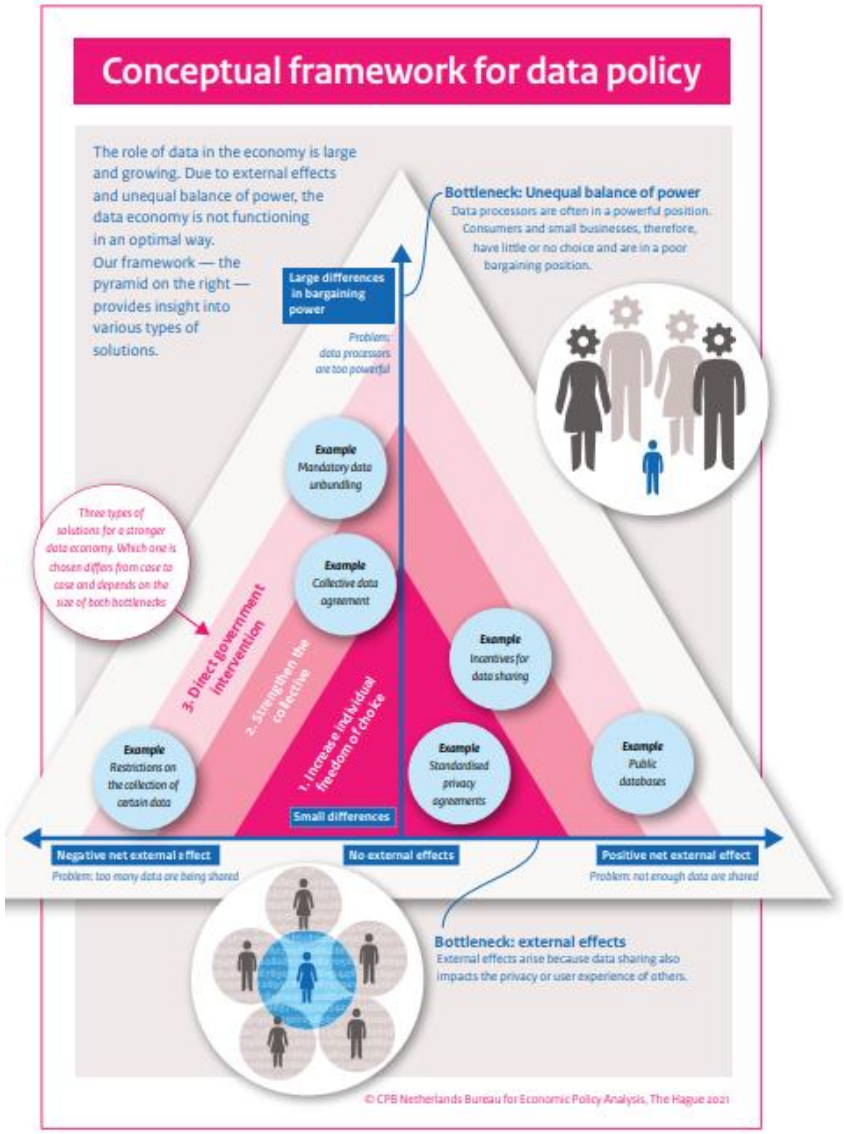
    data breaches undermine confidence in overall data security and willingness to share data, thus affecting other data users and companies may not be charged for such breaches

Information inadequacies

- Competitive markets can only function properly if consumers are sufficiently well informed to evaluate competing products or services
- But information production is costly
- There may be incentives not to disclose

Unequal bargaining power

- Typically, due to problems with
  - competition
  - information
  - coordination
- Particularly when switching costs are high
- Acute problem in data markets – mainly data processors determine, within the legal framework, which personal data are collected and how they are used, rather than the party supplying data



Source: Netherlands Bureau for Economic Policy Analysis

Anti-competitive behavior

- Collusion
- Predatory pricing
- In data markets
  - algorithmic collusion (collusion facilitated through automated systems) ▷ Algorithms and Collusion (OECD)
  - personal pricing algorithms (e.g., set prices based on location)

Public goods

- Individuals do not have the incentives to maintain and improve public goods or goods that
  - non-excludable individuals cannot be excluded from use or could be enjoyed without paying for it AND
  - non-rivalrous use by one individual does not reduce availability to others or the goods can be effectively consumed simultaneously by more than one person
- Consider a public meadow, any farmer can bring their cattle to the meadow, but end result could be that the meadow is destroyed
- Or consider clean air
- The tragedy of the commons

Data as public good?

- Non-excludable, in principle
- Non-rivalrous, in principle
- Subject to the commons problem?
- Property rights are the answer?
- Anti-commons problem?

Why regulate ?

- Social rationales for regulating
  - Dealing with redistribution issues
  - Also, continuity and availability of service (hospital, transportation)
  - Accountability

Why regulate?

- *Rule by Data: The End of Markets?* (Pistor)
- Data not as an object of exchange and regulation but rather as a source and means of control
- Characteristic of public goods rather than private goods
- Can be harvested in large amounts – data processors only need to take them
- This gives them (predictive) power
- They don't really need property rights protections (have technical means)
- Still, they seek such protections (IP etc.)

Why regulate?

- Should we allocate these rights to producers?
- Legally, quite difficult (compare *Regulating Data as Property*, Ritter and Mayer)
- Need to be defined and managed collectively
- Problem is lack of accountability (for ability to influence)

Rights-based rationales

- Protection of human rights
  - freedom of expression
  - right to privacy,
  - right to work and education,
  - prevention of discrimination
  - etc.

Generally enshrined in international and constitutional law, see e.g., Universal Declaration of Human Rights by the United Nations, US Constitution, Constitution of the Kingdom of the Netherlands